

Paper Report: Outlier Robust Mean Estimation with Subgaussian Rates via Stability

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In algorithmic statistics, there has been much interest in the past decade in finding high-dimensional mean estimators that (i) are robust to adversarial corruption, (ii) achieve subgaussian rates, and (iii) are computationally efficient. A well-known one-dimensional solution to this problem is the median of means. Various high-dimensional generalizations of the median have yielded estimators which satisfy (i) and (ii) but not (iii) [Kam25].

Some robust algorithms satisfying (i) and (iii) are known to do well under a particular *stability* condition on the dataset, which is defined below. It was an open question whether there exists an estimator satisfying all three, and this paper answers in the affirmative. This paper shows that previously obtained rates for these robust and efficient algorithms were suboptimal, and that they can in fact attain subgaussian rates [DKP20].

We summarize their main theorems, some technical lemmas, and sketches of their proofs.

Notation. Let $S \subset \mathbb{R}^d$. Let $\mu_S = \frac{1}{|S|} \sum_{x \in S} x$ and $\bar{\Sigma}_S = \frac{1}{|S|} \sum_{x \in S} (x - \mu)(x - \mu)^T$ denote the mean and second moment matrix w.r.t. the uniform distribution on S . Let $\|v\|$ denote the ℓ_2 -norm and let $r(M) = \text{tr}(M)/\|M\|$.

For $0 < \epsilon < 1$, an ϵ -corrupted set is a dataset T such that an adversary can inspect the dataset and corrupt at most ϵn data points. This is the so-called “strong contamination” model in robust statistics. ϵ is typically given as a parameter to the robust algorithm.

Definition 1 ((ϵ, δ) -stability). Fix $0 < \epsilon < 1/2$ and $\delta > \epsilon$. A finite set $S \subset \mathbb{R}^d$ is (ϵ, δ) -stable with respect to mean $\mu \in \mathbb{R}^d$ and σ^2 if for every $S' \subset S$ with $|S'| \geq (1 - \epsilon)n$, the following hold:

- (a) $\|\mu_{S'} - \mu\| \leq \sigma\delta$
- (b) $\|\bar{\Sigma}_{S'} - \sigma^2 I\| \leq \sigma^2 \delta^2 / \epsilon$

In this theorem, the authors prove that as long as the covariance is finite, w.p. $\geq 1 - \exp(-cn\epsilon')$, there exists a large stable subset, and it therefore follows that any stability-based robust algorithm can achieve a near-subgaussian rate.

Theorem 1. *Fix $0 < \tau < 1$. Let S consist of n i.i.d. samples from a distribution on $\mathbb{R}^d$ with mean μ and covariance Σ . Let $\epsilon' = \Theta\left(\frac{\log(1/\tau)}{n} + \epsilon\right) \leq c$ for sufficiently small constant $c > 0$. Then, w.p. $\geq 1 - \tau$, there exists a subset $S' \subset S$ such that $|S'| \geq (1 - \epsilon')n$ and S' is $(2\epsilon', \delta)$ -stable w.r.t. μ and $\|\Sigma\|$, where $\delta = O\left(\sqrt{\frac{r(\Sigma) \log r(\Sigma)}{n}} + \sqrt{\epsilon} + \sqrt{\frac{\log(1/\tau)}{n}}\right)$.*

Sketch of proof. The authors first show that, to prove that S' is $(\Theta(\epsilon), \delta')$ -stable, it suffices to show Properties (a) and (b) of Definition 1 for $\delta' = O(\delta + \sqrt{\epsilon})$. Then, to prove the theorem, it remains to show that, w.p. $1 - 2\exp(-cn\epsilon)$, there exists a large enough subset satisfying those two properties.

The discrete problem of finding a suitable subset is relaxed to the continuous problem of finding a distribution on points. Denote

$$\Delta_{n,\epsilon} = \{w \in \mathbb{R}^d : 0 \leq w_i \leq \frac{1}{(1-\epsilon)n}; \sum_{i=1}^n w_i = 1\}$$

the convex hull of distributions on $S' \subset S$ with $|S'| \geq (1 - \epsilon)n$. It is shown in a lemma that one can recover a desired subset S' from w .

Their Lemmas 2.2-2.4 prove that, assuming $\|x_i - \mu\| = O(\sqrt{\text{tr}(\Sigma)/\epsilon})$, we have that (b) holds w.p. $1 - 2\exp(-cn\epsilon)$ with $\delta = O(\sqrt{\text{tr}(\Sigma) \log r(\Sigma)}/n + \sqrt{\epsilon})$. To do this they use minimax duality to show

$$\min_{w \in \Delta_{n,\epsilon}} \|\bar{\Sigma}_w - \sigma^2 I\| \leq 1 + \max_{M \in \mathcal{M}} \min_{w \in \Delta_{n,\epsilon}} \left\langle \sum_{i=1}^n w_i (x_i - \mu)(x_i - \mu)^T, M \right\rangle, \quad (1)$$

which they upper bound by an empirical process. They then approximate it by a truncated process that is upper bounded using Talagrand's concentration inequality, contraction and symmetrization of Rademacher averages, and the matrix-Bernstein inequality.

Their Lemmas 2.5-2.8 prove that, given the u^* which attains (1) and thus satisfies (b), we can remove an additional $O(\epsilon)$ -fraction of probability mass from u^* to obtain a $w \in \Delta_n$ s.t. (a) also holds. To do this, they simplify the domain constraint on w and obtain a bounded empirical process, which they bound using Talagrand's to show that $\|\mu_{w^*} - \mu\| \leq \delta$.

By combining previous lemmas, the theorem is proven in the case that $\|x_i - \mu\| = O(\sqrt{\text{tr}(\Sigma)/\epsilon})$. As for the general case, they use a Chernoff bound to show that given samples from an unbounded distribution, there exists C' large enough s.t. w.p. $\geq 1 - \exp(-cn\epsilon'/C'^2)$, more than $(1 - \epsilon'/2)n$ points lie within $C'\sqrt{\text{tr}(\Sigma)/\epsilon'}$ of μ for a $c > 0$. The bounded case then applies, and with a slight adjustment using triangle inequality and Cauchy-Schwarz, w.p. $\geq 1 - \tau$, there exists a $(C\epsilon', \delta)$ -stable set of size $\geq (1 - \epsilon')n$ w.r.t. μ and Σ .

Proposition 1. *Let T be an ϵ -corrupted set of n samples from a distribution in $\mathbb{R}^d$ with mean μ and covariance Σ . Let $\epsilon' = \Theta\left(\frac{\log(1/\tau)}{n} + \epsilon\right) \leq c$ be given for a constant $c > 0$. Then any stability-based algorithm on input T and ϵ' efficiently computes $\hat{\mu}$ s.t., w.p. $\geq 1 - \tau$, we have*

$$\|\hat{\mu} - \mu\| = O\left(\sqrt{\frac{\text{tr}(\Sigma) \log r(\Sigma)}{n}} + \sqrt{\|\Sigma\| \epsilon} + \sqrt{\frac{\|\Sigma\| \log(1/\tau)}{n}}\right)$$

This follows from Theorem 1.

The authors further prove that in order to remove the $\sqrt{\log r(\Sigma)}$ factor, it suffices to randomly partition the sample into buckets and run any stability-based algorithm on the means of the buckets:

Proposition 2. *Consider the setting of Proposition 1. Let $k = \lfloor \epsilon' n \rfloor$ and $\{z_1, \dots, z_k\}$ be the means of k random buckets of the corrupted set T . Then any stability-based algorithm on input T and ϵ' efficiently computes $\hat{\mu}$ s.t. w.p. $\geq 1 - \tau$, we have*

$$\|\hat{\mu} - \mu\| = O\left(\sqrt{\frac{\text{tr}(\Sigma)}{n}} + \sqrt{\|\Sigma\| \epsilon} + \sqrt{\frac{\|\Sigma\| \log(1/\tau)}{n}}\right)$$

Thus, any stability-based algorithm can be used as the aggregator in the median-of-means paradigm to obtain subgaussian rates efficiently. This produces an estimator that is (i) robust to adversarial corruption, (ii) achieves subgaussian rates, and (iii) is efficiently computable.

To prove Proposition 2, the authors prove the following theorem relating the median-of-means principle and stability:

Theorem 2. *Let T be an ϵ -corrupted version of the set S , where S consists of n i.i.d. points from a distribution P with mean μ and covariance Σ . Set $\epsilon' = O(\epsilon \log(1/\tau)/n)$ and set $k = \lfloor \epsilon' n \rfloor$. Let $T_k = \{z_1, \dots, z_k\}$ be the set of k points obtained by the median-of-means preprocessing on the set T . Then, w.p. $1 - \tau$, T_k is a 0.01-corruption of a set S_k s.t. there exists a $S'_k \subset S_k$ s.t. $|S'_k| \geq 0.95k$ and S'_k is $(0.1, \delta)$ -stable w.r.t. μ and $k\|\Sigma\|/n$, where $\delta = O(\sqrt{r(\Sigma)}/n + 1)$.*

Sketch of proof. First, the (highly probable) existence of a stable subset is shown for S_k , the means of the uncorrupted set. To find S'_k with Property (b), they again find $\min_{w \in \Delta_{n, \epsilon}} \|\bar{\Sigma}_w - \sigma^2 I\|$. The minimum second moment achievable by sets of size $\geq 0.99k$ is bounded in a manner similar to (1) using minimax duality and an existing theorem from the literature that upper bounds $(z_i - \mu)^T M(z_i - \mu)$ w.h.p. The second moment is then used to bound the first moment, yielding w^* , and, in turn, an S'_k with Property (a).

It remains to show that T_k is a corrupted version of S_k . In the worst case, the adversary corrupted one entry in each ϵn bucket, so $|T_k \cap S_k| \geq k - \epsilon n$. Since $k - \epsilon n = \lfloor \epsilon' n \rfloor - \epsilon n \geq 0.99k = 0.99 \lfloor \epsilon' n \rfloor$ for ϵ' large enough, T_k is at most a (0.01) -contamination of S_k .

Lastly, it is shown that in the case of identity covariance and bounded k moments, one can obtain tighter bounds than Theorem 1.

Theorem 3. *Let S be a multiset of n i.i.d. samples from a distribution on $\mathbb{R}^d$ with mean μ and covariance $\Sigma = I$, and bounded central moment σ_k , for some $k \geq 4$. Let $\epsilon' = \Theta(\log(1/\tau)/n + \epsilon) \leq c$ for sufficiently small $c > 0$. Then, w.p. $\geq 1 - \tau$, there exists a subset $S' \subset S$ s.t. $|S'| \geq (1 - \epsilon')n$ and S' is $(2\epsilon', \delta)$ -stable w.r.t. μ and $\sigma^2 = 1$, where $\delta = O(\sqrt{d \log d/n} + \sigma_k \epsilon^{1-\frac{1}{k}} + \sigma_4 \sqrt{\log(1/\tau)/n})$.*

The proof is nearly the same as that of Theorem 1. Whereas, previously, only an upper bound on the second moment matrix was needed, we now need a sharp lower bound on the minimum eigenvalue of $\bar{\Sigma}_{S_1}$ over all stable subsets. The proof now uses the variational characterization of the spectral norm and a lemma which applies Jensen's, Hölder's, and Markov's inequalities.

Much like Proposition 1, it follows that a stability-based algorithm on a distribution with the additional constraints can also attain better bounds.

Proposition 3. *Let T be an ϵ -corrupted set of n points from a distribution on $\mathbb{R}^d$ with mean μ and covariance $\sigma^2 I$ and k -th bounded central moment σ_k , $k \geq 4$. Let $\epsilon' = \Theta(\log(1/\tau)/n + \epsilon) \leq c$ be given for some $c > 0$. Then any stability based algorithm, on input T and ϵ' , efficiently computes $\hat{\mu}$ s.t. w.p. $\geq 1 - \tau$, we have $\|\hat{\mu} - \mu\| = O(\sigma(\sqrt{d \log d/n} + \sigma_k \epsilon^{1-\frac{1}{k}} + \sigma_4 \sqrt{\log(1/\tau)/n}))$.*

This paper proved that previously proposed efficiently computable robust estimators did in fact attain near-subgaussian rates, answering an open question. Moreover, with simple median-of-means preprocessing, the log factor can be removed. Any stability-based algorithm can thus be used to aggregate the bucket means of a corrupted sample (from a distribution with finite covariance) with subgaussian rates. This was done by proving stability properties that hold w.h.p. about samples from distributions with finite covariance.

References

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